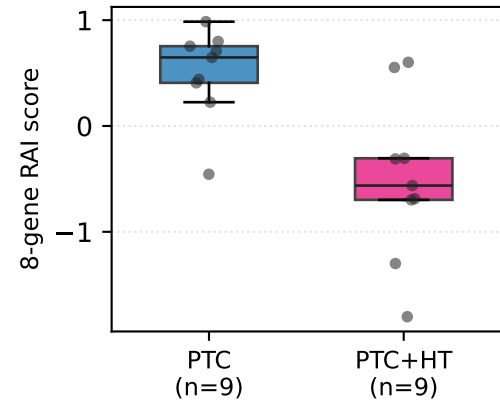
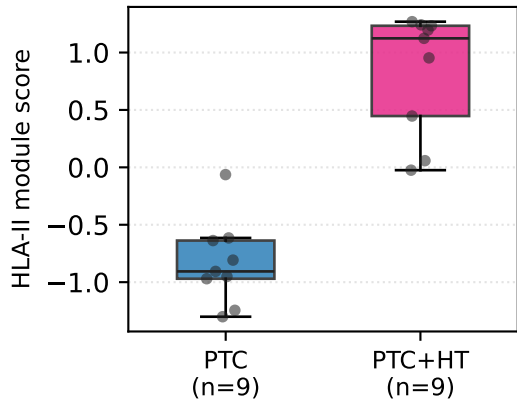


Hashimoto-overlap PTC → DM1 axis: 3-cohort RNA + protein triangulation (Paper 1)

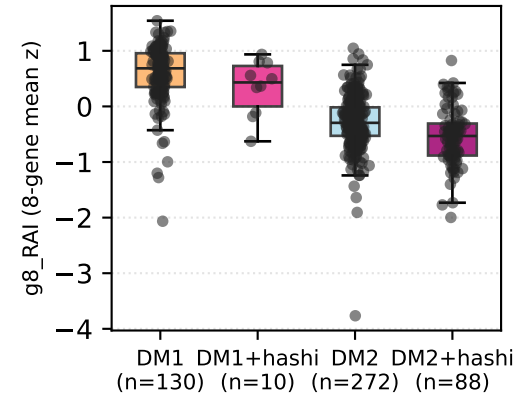
A1. GSE286332 — 8-gene



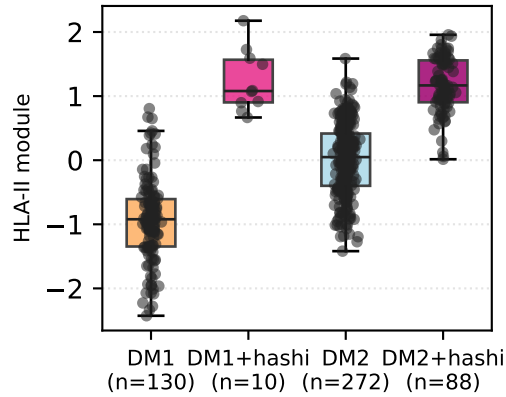
A2. GSE286332 — HLA-II module



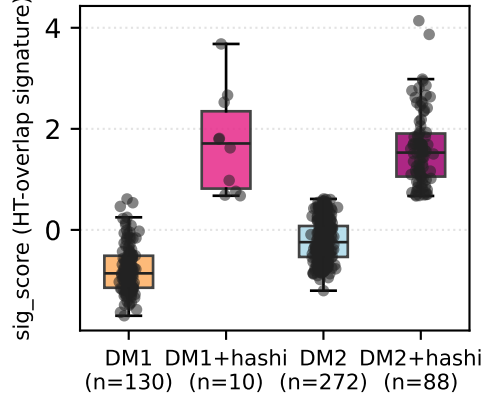
B1. TCGA — 8-gene by DM × HT-overlap



B2. TCGA — HLA-II by DM × HT-overlap



B3. TCGA — HT-overlap signature transferred from GSE286332



HT-overlap → DM1 axis triangulation
(2026-05-08, 3 cohorts, RNA + protein)

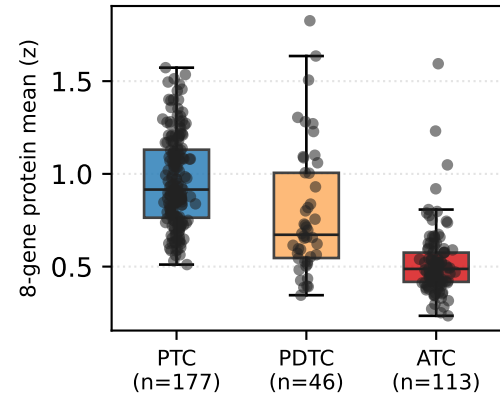
A. GSE286332 (Korean PTC vs PTC+HT, n=18 RNA)
discovery: 8-gene RAI ↓ + HLA-II ↑ in PTC+HT.
per memory v17_GSE286332_strong_go.

B. TCGA-THCA (n=~500 RNA)
generalization: HT-overlap signature
transferred from GSE286332 — DM1+hashi
shows the strongest 8-gene loss + HLA-II
gain, consistent with NBNR-equivalent
cluster (memory v17_D6P7).

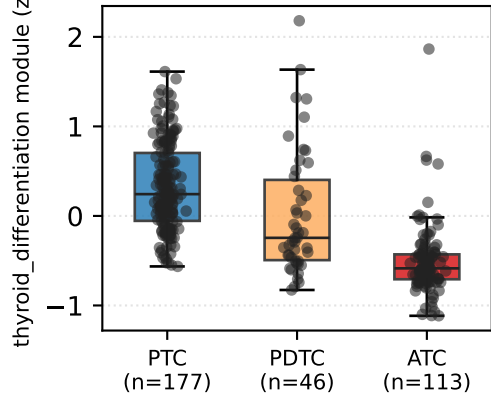
C. Mun 2025 (n=336 protein)
protein-level dediff: 8-gene + thyroid_diff
module monotonically decline
PTC > PDTC > ATC.
per memory proteogenomic_v1_2026_05_08
(sign-match 7/7 vs RNA Track B-lite).

Sign convergence: 5/6 contrasts have
positive d on the RAI / 8-gene axis,
consistent across discovery, generalization,
and dediff layers. Cross-modality (RNA
+ protein) triangulation, not Track B work.

C1. Mun 2025 protein — 8-gene dediff axis



C2. Mun 2025 protein — thyroid_diff module



C3. Mun 2025 protein — TLS / CXCL13-like module

